MingYu Zhao

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EDUCATION

Rutgers University, Graduate School of Arts and Sciences — New Brunswick, NJ
Ph.D. in Computer Science — Expected May 2029

University of Science and Technology of China — Hefei, China
B.Eng. in Artificial Intelligence — Jun 2024

PUBLICATIONS *"*" denotes equal contribution.*

First-author Publications

- **Mingyu Zhao***, S. Lin*, L. Han, X. Yu, Z. Wang, Y. Pang, Y. Zhou, D. N. Metaxas. "Metacognitive Decoding: Efficient Inference for Diffusion Language Agents." *arXiv preprint*, 2026.
- **Mingyu Zhao***, Y. Zhou*, Z. Wang, D. Gu, B. Guo, R. Ye, L. Han, C. Jin, D. N. Metaxas. "M3-Bench: Multi-Modal, Multi-Hop, Multi-Threaded Tool-Using MLLM Agent Benchmark." Submitted to the *NeurIPS*, 2026.
- **Mingyu Zhao**, Z. Yang, Y. Zhou, Z. Xia, C. Jin, X. He, C. Neidle, D. N. Metaxas. "MHB: Multimodal Handshape-aware Boundary Detection for Continuous Sign Language Recognition." Accepted by the *LREC*, 2026.
- **Mingyu Zhao***, Z. Li*, X. Dong, H. Ling, B. Huang. "CAPAA: Classifier-Agnostic Projector-Based Adversarial Attack." Accepted by the *IEEE ICME*, 2025.

Selected Other Publications

- "Lor-VP: Low-rank Visual Prompting for Efficient Vision Model Adaptation." *ICLR 2025.*
- "Large Sign Language Models: Toward 3D American Sign Language Translation." *WACV 2026.*
- "LED: LLM-Enhanced Open-Vocabulary Object Detection without Human-Curated Data Generation." *CVPR 2026.*

RESEARCH EXPERIENCE

Topic: Efficient Inference for Diffusion Language Agents
Instructor: Prof. Dimitris N. Metaxas (Rutgers University)

- Developed a lightweight controller that reduces unnecessary computation in diffusion language agents by dynamically deciding when to finalize tokens and how far to generate, without updating the underlying model.
- Applied **Agentic RL (GRPO)** to train the controller on complete agent interactions, rewarding correct task completion with lower computational cost.
- Reduced inference FLOPs by **24.6–43.1% across five benchmarks** compared with the same model without the controller, while improving accuracy on four benchmarks.

Topic: Multi-Hop, Multi-Threaded MLLM Agent Evaluation and Optimization — *Oct. 2025 ∼ Apr. 2026*
Instructor: Prof. Dimitris N. Metaxas (Rutgers University)

- Built an end-to-end MCP benchmark generation and correctness validation framework; designed an MCP benchmark suite with 200+ tasks involving visual grounding and cross-tool dependencies, each with a standardized trajectory.
- Proposed a structure-aware tool-call evaluation metric using **sentence-encoder** embeddings and **similarity-bucketed Hungarian matching**; quantitatively evaluated GPT, Gemini, Claude, and **10+** MLLM models across the benchmark, with qualitative and quantitative analyses of budget and error types to characterize tool-call failure modes.
- Performed **supervised fine-tuning** on LLaDA 1.5 using generated tool-call data from multiple datasets, integrated a reasoning module, and optimized the model with **reinforcement learning**, improving specified tool-call task performance by **32%** over the base LLaDA 1.5 tool-calling baseline under **LLM judge** scores and accuracy metrics.

Topic: Multi-Modal Video Temporal Boundary Detection — *Jan. 2025 ∼ Dec. 2025*
Instructor: Prof. Dimitris N. Metaxas (Rutgers University) and Prof. Carol Neidle (Boston University)

- Designed a spatio-temporal convolution-based segmentation module with velocity and acceleration features from skeleton sequences, improving **temporal boundary detection** F1 score by around **10%** over previous methods.
- Integrated handshape information by pretraining a handshape classifier over canonical categories and fusing the classifier's learned hidden representations into the segmentation stream via **cross-attention** with a gating mechanism.
- Built an **end-to-end** continuous sign language recognition pipeline by combining boundary detection with sign recognition, improving boundary detection F1 score by nearly **20%** overall; for signs with well-localized boundaries, continuous recognition performance was within only **10%** of isolated-sign recognition.

Topic: Classifier-Agnostic Projector-Based Adversarial Attack — *July. 2024 ∼ Dec. 2024*
Instructor: Prof. HaiBin Ling (Stony Brook University) and Prof. BingYao Huang (Southwest University)

- Proposed a pose-agnostic projector-based attack algorithm using Grad-CAM++ to focus perturbations on salient physical regions, mitigating the effects of camera-pose changes and improving visual stealthiness by around **20%**.
- Designed a novel optimal loss function to generate projector-based adversarial attacks across multiple classifiers with **negligible performance degradation** compared with classifier-specific attacks.
- Validated the effectiveness of the proposed CAPAA by building a camera-projector system in the real world and by evaluating the attack success rate and stealthiness on 10 sets of ablation experiments.

INDUSTRY PROJECT

Topic: ASL Pal *Sep. 2025 ~ Present*

Leads: Beryl Breindel (DawnSignPress) and Prof. Dimitris N. Metaxas (Rutgers University)

- Fine-tuned a state-of-the-art isolated American Sign Language (ASL) recognition model on the company's proprietary ASL dataset, achieving **85%** recognition accuracy and outperforming the original model.
- Engineered the backend inference pipeline with model compression, parallel execution, model preloading, and other technics, reducing model runtime from **5 min** to **15 sec**.
- Integrated the models into the web platform with the engineering team and validated the system through user testing.

TECHNICAL STRENGTHS

Skills: Python, PyTorch, Bash, C/C++, AI-assisted IDE, Agentic Coding, Linux, MCP, Git, Flask, Slurm, LaTeX.